

AI-Driven Vulnerability Analysis

1. The Vulnerability Management Problem at Scale

A typical enterprise running 10,000 endpoints may generate 50,000 to 100,000 vulnerability findings per month from traditional CVSS-scored scanners. If the vulnerability management team has five analysts, each analyst would need to triage 2,000 to 4,000 findings per week — an impossible manual workload. The documented result, consistent across Tenable's annual Vulnerability Intelligence Reports, is that organizations patch a small fraction of discovered vulnerabilities and frequently apply the wrong patches first.

AI-driven vulnerability analysis changes the problem from 'how do we process everything?' to 'what do we actually need to fix first?' This is not primarily a scanning problem — modern scanners are already fast and comprehensive. It is a prioritization and context problem. An AI system that understands asset criticality, exploit availability, network exposure, and business process dependencies can produce an actionable ranked work order from 50,000 findings in seconds. That same ranking would take a manual analyst team weeks to produce with comparable accuracy.

2. From CVSS to Risk-Based Prioritization

CVSS (Common Vulnerability Scoring System) scores vulnerabilities on technical characteristics: attack vector, complexity, privileges required, user interaction, and impact. A CVSS 9.8 appears critical on paper, but if the affected component exists only on an air-gapped development server, the actual organizational risk is low. Conversely, a CVSS 6.5 vulnerability in an internet-facing authentication endpoint with active exploitation in the wild represents far greater real risk than its score suggests.

Priority Signal	What It Captures	Example Tool That Uses It
CVSS base score	Technical severity of the vulnerability (attack vector, complexity, impact)	All scanners — Nessus, Qualys, Rapid7
Exploit availability	Is working exploit code publicly available? (Exploit-DB, Metasploit, GitHub)	Tenable VPR, Qualys TruRisk, Rapid7 InsightVM
Active exploitation status	Is this CVE being exploited by known threat actors right now?	CISA KEV catalog, threat intelligence feeds
Asset criticality	How critical is the affected system to business operations?	Tenable One, Qualys VMDR with asset tagging
Network exposure	Is the asset internet-facing? Reachable from untrusted network segments?	Wiz, Orca Security (attack path analysis)
Patch stability	Has the patch been validated as stable? Are there known regression risks?	Ivanti Neurons for Patch Intelligence

Core Concept: CVSS measures technical severity. Risk-based prioritization measures organizational exposure. A CVSS 6.5 on an internet-facing system with active exploitation in the wild is higher priority than a CVSS 9.8 on an isolated internal server. The exam tests this distinction frequently.

3. Continuous vs. Point-in-Time Scanning

Traditional vulnerability scanning was periodic — weekly or monthly runs using Nessus, OpenVAS, or Qualys. The problem: attack surfaces change continuously. A developer pushes a container image with a vulnerable base layer on Tuesday; the CVE for that base layer is disclosed on Tuesday afternoon; the next scheduled scan runs Friday. That is a 72-hour window of unknown exposure during which the organization cannot detect, respond to, or report on a vulnerability that may already be under active exploitation.

AI-driven platforms address this with persistent agents and API integrations that provide near-real-time asset inventory and continuous assessment. Cloud Security Posture Management (CSPM) tools extend this to cloud resource configurations — flagging misconfigured S3 buckets, overly permissive IAM policies, or exposed databases as they are created, not during the next scheduled scan. The SecAI+ exam expects candidates to recognize continuous monitoring as a distinct capability from periodic scanning, and to know which NIST framework controls mandate it (NIST SP 800-53 CA-7).

4. Real-World Incident: ConnectWise ScreenConnect Exploitation Window (2024)

Real-World Incident — ConnectWise ScreenConnect CVE-2024-1709 (March 2024): A critical authentication bypass vulnerability (CVSS 10.0) in ConnectWise ScreenConnect was publicly disclosed on a Tuesday. Working exploit code appeared on GitHub within hours of disclosure. By the following Monday, threat intelligence sources had documented active exploitation by ransomware groups including Black Basta and LockBit affiliates — making this one of the fastest weaponizations of a disclosed vulnerability on record. Organizations with AI-driven continuous scanning and threat intelligence integration received automated alerts on Tuesday identifying affected instances and linking to active exploitation reports. Their ticketing systems received auto-created remediation work orders the same day. Organizations relying on weekly scheduled scans learned about their exposure during the Friday scan — by which time attackers had already had four days to establish footholds.

This incident is a clean illustration of why continuous monitoring with threat intelligence integration is qualitatively different from scheduled scanning. The vulnerability disclosure window — the period between public disclosure and organizational patch deployment — is precisely where AI-driven continuous monitoring provides its highest value. CISA's Known Exploited Vulnerabilities (KEV) catalog was created specifically to close this gap, listing CVEs with confirmed active exploitation that organizations should prioritize above all others regardless of CVSS score.

5. AI-Assisted Fuzzing for Zero-Day Discovery

Traditional vulnerability scanners identify known CVEs — they are entirely blind to vulnerabilities that have not yet been disclosed. AI-assisted fuzzing generates millions of malformed inputs and monitors applications for unexpected behavior (crashes, memory corruption, unexpected output) that indicates an undisclosed vulnerability. This approach is used offensively by researchers to discover zero-days before attackers do.

Google's OSS-Fuzz project uses AI-guided fuzzing to continuously test critical open-source projects including OpenSSL, SQLite, and the Linux kernel. It has discovered thousands of vulnerabilities that would otherwise have remained undisclosed until an attacker found them independently. Microsoft's Project OneFuzz provides similar capabilities for Windows ecosystem software. Trail of Bits' Echidna targets Solidity smart contracts. The MITRE ATLAS v2.1 framework recognizes AI-Enabled Vulnerability Discovery (AML.T0044) as both a defensive research capability and an offensive risk — adversaries can use the same fuzzing infrastructure to find zero-days for offensive exploitation.

6. Cloud Posture Analysis and Attack Path Correlation

Misconfigurations are the leading cause of cloud security incidents, ahead of vulnerability exploitation, according to the 2024 Verizon Data Breach Investigations Report. AI-driven CSPM tools audit cloud configurations continuously against security benchmarks including CIS Benchmarks, AWS Well-Architected Framework, Azure Security Benchmark, and Google Cloud Security Foundations Guide.

The differentiating capability of advanced CSPM tools (Wiz, Prisma Cloud, Orca Security) is attack path analysis: correlating individual misconfiguration findings to surface compound risk. A standalone finding that an S3 bucket is publicly readable has medium severity. When the CSPM tool additionally knows that the bucket contains AWS credentials, those credentials have IAM permissions to modify production infrastructure, and that production infrastructure hosts the payment processing system, the attack path has critical business impact that no single-point check would reveal. Wiz refers to this as the 'security graph' — a relationship model of all cloud resources and their connections.

7. Contextual Asset Understanding in AI Vulnerability Platforms

The NIST AI RMF 1.0 MAP function requires AI systems in security applications to demonstrate contextual understanding — the ability to account for the environment in which a finding exists, not just the technical characteristics of the finding itself. A vulnerability management platform that treats all CVSS 7.0 findings identically regardless of whether they exist in a DMZ web server or an isolated test environment produces recommendations with no operational value.

Tenable's Vulnerability Priority Rating (VPR) and Qualys TruRisk both implement contextual weighting by combining technical severity with threat intelligence, asset criticality scores, and network exposure information. The output is a ranked list where organizational risk — not abstract technical severity — determines priority. This contextual weighting is exactly what the MAP function in NIST AI RMF 1.0 is designed to ensure: that AI systems understand and account for the deployment context of their outputs.

8. Red Team Perspective: What Automated Scanners Consistently Miss

Understanding scanner limitations is essential for calibrating organizational confidence in automated vulnerability programs:

- **Business logic flaws:** scanners cannot understand that an insurance claims system should not allow a claimant to approve their own submission. This requires understanding the application's intent, not its code structure.
- **Authentication weaknesses requiring session state:** vulnerabilities that emerge from interactions across multiple HTTP requests and user roles cannot be identified by static scanning tools.
- **Second-order vulnerabilities:** data stored in one form (round 1) and processed unsafely in a separate code path (round 2) is frequently missed by scanners that analyze individual operations in isolation.
- **Race conditions:** TOCTOU vulnerabilities require modeling concurrent execution, which static and most dynamic scanners do not do.
- **Custom application-specific bugs:** scanners excel at known CVEs in recognized libraries but are substantially less effective at finding vulnerabilities in bespoke application code.

The responsible posture: treat vulnerability scanner output as a necessary minimum baseline, not comprehensive security validation. Manual penetration testing (covered in the next lecture) complements automated scanning by specifically targeting the categories that automation misses. NIST SP 800-115 (Technical Guide to Information Security Testing and Assessment) recommends combining automated scanning with human-led testing precisely because each method has distinct detection capabilities.

9. AI-Assisted Patch Prioritization and Intelligence

Finding a vulnerability is step one. The operationally important decision is what to patch today, this week, or next quarter. AI-assisted patch intelligence incorporates multiple signals that automated scanners do not traditionally consider:

- Active exploitation status (CISA KEV catalog, threat intelligence feeds)
- Patch stability and quality (has this patch caused known regressions or system instability?)
- Business process dependencies (can this system be rebooted during available maintenance windows?)
- Patch prerequisites (does this patch require an OS version the system is not currently running?)
- Affected asset criticality and business impact classification

Ivanti Neurons for Patch Intelligence and Microsoft Endpoint Configuration Manager's AI-assisted patch scoring incorporate these signals to produce a ranked work order — not a flat list of everything that eventually needs patching. The practical impact: patching teams apply effort proportional to actual organizational risk rather than CVSS score order, which demonstrably improves security outcomes per patching cycle.

10. Compliance Framework Requirements for Vulnerability Management

Several compliance frameworks impose specific vulnerability management requirements that AI-driven tools help satisfy:

Framework / Requirement	Obligation	AI Tool Contribution
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PCI-DSS v4.0 Req 11.3	Internal and external vulnerability scanning at least quarterly and after significant changes	Continuous scanning satisfies and exceeds frequency requirement; auto-documentation provides audit evidence
NIST SP 800-53 CA-7	Continuous monitoring of information system security controls	AI platforms provide real-time monitoring and automated alerting that satisfies CA-7 implementation
HIPAA 164.308(a)(8)	Periodic evaluation to verify technical and non-technical safeguards remain appropriate	Continuous scanning with configuration drift detection supports HIPAA evaluation requirements
FedRAMP	Continuous monitoring with monthly vulnerability scanning, scanning after significant changes	AI-driven continuous monitoring satisfies FedRAMP ConMon frequency requirements

Important compliance nuance for the exam: meeting the scan frequency requirement (e.g., quarterly per PCI-DSS) is not the same as having an effective vulnerability management program. Regulators are increasingly scrutinizing remediation velocity — how quickly critical findings are addressed — not merely whether scans occurred. AI-generated reporting that tracks mean time to remediation (MTTR) for critical findings provides the evidence regulators are looking for.

11. Responsible AI Deployment: False Positive and False Negative Trade-offs

Overly sensitive AI scanners generate false positives — findings that are not real vulnerabilities. Overly conservative scanners produce false negatives — real vulnerabilities that are missed. Both have significant operational costs. False positives burn analyst time and destroy credibility with the engineering teams that receive remediation tickets, leading to ticket backlogs and deprioritization. False negatives leave real vulnerabilities undetected and unexploited until an attacker finds them.

NIST AI RMF 1.0 MEASURE function: AI performance in security applications must be tracked and reported continuously, not assumed to be accurate after initial deployment. The recommended calibration approach is a feedback loop: measure false positive rates through analyst review of confirmed findings, measure false negative rates through periodic red team testing or third-party assessments, and adjust model thresholds and configuration accordingly. This is not a one-time deployment activity — it is an ongoing operational requirement.

Key Terms Glossary

Term	Definition
CVSS (Common Vulnerability Scoring System)	A framework that assigns technical severity scores to vulnerabilities based on attack vector, complexity, privileges required, user interaction, and impact.
Vulnerability Priority Rating (VPR)	Tenable's proprietary risk-based scoring that combines CVSS with threat intelligence, exploit availability, and asset context to produce organizational risk scores.
TruRisk	Qualys's risk-based vulnerability scoring system incorporating asset criticality, threat intelligence, and network exposure alongside

	technical severity.
CSPM (Cloud Security Posture Management)	AI-driven tools that continuously audit cloud resource configurations against security benchmarks and surface misconfigurations and attack paths in real time.
CISA KEV (Known Exploited Vulnerabilities)	CISA's authoritative catalog of CVEs confirmed to be actively exploited by threat actors, mandating priority remediation for federal agencies.
Attack path analysis	CSPM capability that correlates individual findings across cloud resources to identify compound risk chains that lead to critical business impact.
AI-assisted fuzzing	Using ML-guided input generation to discover undisclosed vulnerabilities in software by monitoring for unexpected behavior (crashes, memory corruption) during testing.
OSS-Fuzz	Google's continuous AI-guided fuzzing infrastructure for critical open-source projects, which has discovered thousands of vulnerabilities before public disclosure.
NIST SP 800-53 CA-7	Continuous Monitoring control requiring ongoing monitoring of information system security controls and organizational risk posture.
MTTR (Mean Time to Remediate)	Average time from vulnerability discovery to confirmed remediation; the key metric regulators use to evaluate vulnerability management program effectiveness.

Quick Revision Checklist

- CVSS = technical severity; VPR/TruRisk = organizational risk (adds exploit availability, asset criticality, network exposure).
- Continuous scanning (agents + API) vs. periodic scanning: 72-hour disclosure window example — organizations without continuous monitoring missed ConnectWise exploitation window.
- CISA KEV catalog: authoritative list of actively exploited CVEs; always higher priority than CVSS score alone.
- CSPM tools (Wiz, Prisma Cloud, Orca) provide attack path analysis — correlating individual misconfigurations to surface compound critical risks.
- AI fuzzing (OSS-Fuzz, OneFuzz) discovers zero-days before attackers; MITRE ATLAS AML.T0044 recognizes this as both defensive and offensive capability.
- NIST AI RMF 1.0 MAP function: AI vulnerability tools must demonstrate contextual understanding of deployment environment, not just technical scoring.
- Scanners miss: business logic flaws, TOCTOU, second-order injection, custom application bugs — manual pen testing remains necessary.
- Patch prioritization: CISA KEV status, patch stability, maintenance window availability, asset criticality — not CVSS order.
- Compliance: PCI-DSS v4.0 Req 11.3 (quarterly scans), NIST SP 800-53 CA-7 (continuous monitoring), FedRAMP ConMon (monthly scans).

- NIST AI RMF MEASURE function: AI scanner performance must be continuously calibrated via feedback loops — false positive rate from analyst review, false negative rate from red team testing.

10 Exam-Based MCQs — AI-Driven Vulnerability Analysis

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. Scenario: A security operations team at a financial institution learned about the ConnectWise ScreenConnect critical authentication bypass (CVE-2024-1709) through a vendor advisory two days after disclosure, by which time ransomware groups had already established footholds in organizations running unpatched instances. The team runs weekly vulnerability scans. Their CISO has asked them to ensure that future critical disclosures result in detection and ticketing within four hours of public disclosure. Which approach should the security team implement to MOST effectively reduce the time between vulnerability disclosure and detection of affected systems?

- A) Increase scheduled scan frequency from monthly to weekly
- B) Deploy AI-driven continuous monitoring with threat intelligence integration and automated ticketing
- C) Subscribe to vendor security advisories and manually review them each morning
- D) Implement a SIEM that aggregates vulnerability scanner logs for correlation

Answer: B **Explanation:** B is correct. The ConnectWise scenario demonstrates that the disclosure-to-exploitation window can be less than 24 hours. Only continuous monitoring with threat intelligence integration — which identifies affected assets and links them to active exploitation intelligence in real time — can reliably detect exposure within that window. A is wrong — weekly scans still leave a 7-day gap; the scenario requires hours, not days. C is wrong — manual advisory review cannot scale to the speed or volume required, and does not automatically identify which organizational assets are affected. D is wrong — a SIEM aggregates logs from existing scanners but does not replace the need for continuous scanning; it cannot surface findings from a scheduled scanner that hasn't run yet.

Q2. Scenario: A vulnerability management team receives their weekly scan results. Among 3,000 findings, two are flagged for review: a CVSS 9.8 SQL injection in an internal database server accessible only from the development VLAN with no public-facing exposure and no known exploits, and a CVSS 7.2 authentication bypass in an internet-facing web application that appears on the CISA KEV catalog as actively exploited by ransomware groups. Which prioritization approach would MOST accurately identify the HIGHEST-RISK finding in this environment?

- A) Patch the CVSS 9.8 database server finding first because it has the highest technical severity score
- B) Patch the CVSS 7.2 web application finding first because it is internet-facing with active exploitation confirmed
- C) Patch all findings in CVSS score order, highest to lowest
- D) Patch findings alphabetically by CVE identifier to ensure consistent process

Answer: B **Explanation:** B is correct. Risk-based prioritization combines technical severity with

organizational context: exploit availability, active exploitation status, and asset network exposure. A CVSS 7.2 finding with confirmed active exploitation on an internet-facing system presents greater real organizational risk than a CVSS 9.8 finding on an isolated internal server with no known exploits. A is wrong — CVSS 9.8 reflects technical severity but not organizational exposure; the isolated database presents lower actual risk despite the higher score. C is wrong — patching in CVSS score order without context is the exact problem AI-driven vulnerability analysis is designed to solve. D is wrong — CVE identifier order is alphabetical/numeric and has no relationship to risk.

Q3. Which NIST framework control MOST directly mandates the continuous monitoring approach that AI-driven vulnerability platforms provide?

- A) NIST SP 800-218 PW.7.1 — automated static analysis in development
- B) NIST SP 800-53 CA-7 — continuous monitoring
- C) NIST AI RMF 1.0 MAP 1.6 — contextual risk identification
- D) NIST CSF 2.0 RS.AN-03 — incident analysis

Answer: B **Explanation:** B is correct. NIST SP 800-53 CA-7 (Continuous Monitoring) explicitly requires ongoing monitoring of security controls and organizational risk posture — the control that AI-driven continuous vulnerability monitoring directly implements. A is wrong — SP 800-218 PW.7.1 addresses automated static analysis in the software development process, not production environment vulnerability monitoring. C is wrong — NIST AI RMF 1.0 MAP 1.6 is about understanding the context in which AI systems operate, not a monitoring frequency requirement. D is wrong — RS.AN-03 covers incident response analysis activities, not pre-incident vulnerability monitoring.

Q4. A CSPM tool identifies that an S3 bucket is publicly readable, contains AWS access keys, and those keys have IAM permissions to modify the production database. Which capability of the CSPM tool surfaced this risk?

- A) Continuous scanning — identifying the misconfiguration faster than a scheduled scan would
- B) Attack path analysis — correlating individual findings to identify a compound critical risk chain
- C) Behavioral anomaly detection — flagging unusual access patterns on the S3 bucket
- D) Signature matching — identifying the exposed credentials against known compromised credential patterns

Answer: B **Explanation:** B is correct. Attack path analysis correlates multiple individual findings (public S3 bucket + contains credentials + those credentials have privileged access) to surface a risk chain that no single-finding check would reveal. The S3 bucket alone is medium severity; the attack path to production database modification is critical. A is wrong — continuous scanning is about monitoring frequency, not about correlating findings. C is wrong — behavioral anomaly detection monitors access patterns against baselines; it is not the capability that connected the three distinct findings into an attack path. D is wrong — signature matching identifies known patterns; attack path analysis is relationship modeling, not pattern matching.

Q5. Which vulnerability class is LEAST likely to be discovered by an AI-driven automated vulnerability scanner, even one with contextual understanding?

- A) A CVE in a known version of Apache Tomcat running on a web server
- B) A misconfigured S3 bucket allowing public read access
- C) A business logic flaw allowing a customer to apply discount codes created for internal employees
- D) An unpatched OpenSSL version with a known critical CVE

Answer: C **Explanation:** C is correct. Business logic flaws require understanding the application's intended business purpose — that discount codes created for internal employees should not be usable by customers is a business rule, not a code structure pattern. Automated scanners have no model of business intent and will not flag this. A is wrong — CVEs in known versions of recognized software are exactly what automated scanners are designed to detect. B is wrong — CSPM tools continuously check S3 bucket ACL configurations against CIS Benchmarks and flag public access; this is a standard CSPM finding. D is wrong — known CVEs in detected library versions are the core use case for vulnerability scanners.

Q6. The NIST AI RMF 1.0 MAP function requires AI vulnerability management systems to demonstrate which characteristic?

- A) The ability to scan all network assets simultaneously without performance degradation
- B) Contextual understanding of the deployment environment so outputs reflect actual organizational risk, not just technical severity
- C) Integration with all major SIEM platforms for automated alert ingestion
- D) Support for the CVSS v4.0 scoring standard released in 2023

Answer: B **Explanation:** B is correct. NIST AI RMF 1.0 MAP function requires AI systems to understand and account for the context in which they are deployed. For vulnerability management, this means incorporating asset criticality, network exposure, and business process dependencies into findings — not treating all CVSS 7.0 findings identically regardless of context. A scanner that produces the same recommendation for an internet-facing payment processor and an isolated test environment demonstrates failure of contextual understanding. A is wrong — scanning performance is an engineering requirement, not an AI RMF MAP requirement. C is wrong — SIEM integration is an interoperability requirement, not addressed by the MAP function. D is wrong — CVSS version support is a standards compliance matter unrelated to AI RMF contextual understanding.

Q7. Google's OSS-Fuzz project MOST directly demonstrates which AI security capability described in MITRE ATLAS v2.1?

- A) AML.T0015 — Evading ML-based detection systems
- B) AML.T0044 — AI-Enabled Vulnerability Discovery
- C) AML.T0020 — Poisoning training data to degrade model performance
- D) AML.T0043 — Backdoor ML model

Answer: B **Explanation:** B is correct. MITRE ATLAS v2.1 AML.T0044 (AI-Enabled Vulnerability Discovery) documents the use of AI-guided fuzzing to discover vulnerabilities in software — the exact capability OSS-Fuzz implements. OSS-Fuzz uses ML to guide input generation toward code paths most likely to trigger

vulnerabilities. ATLAS recognizes this as both a defensive research capability and an offensive risk. A is wrong — AML.T0015 covers ML evasion, not vulnerability discovery. C is wrong — AML.T0020 covers training data poisoning. D is wrong — AML.T0043 covers backdooring ML models.

Q8. A CISO asks whether the organization's vulnerability management program is effective. Which metric provides the MOST direct evidence of program effectiveness beyond scan frequency compliance?

- A) Total number of vulnerability findings per scan cycle
- B) Mean time to remediate (MTTR) critical and high findings
- C) Number of scanning licenses deployed across the organization
- D) Percentage of assets covered by the vulnerability scanner agent

Answer: B **Explanation:** B is correct. Regulators and security leadership are increasingly focused on remediation velocity — how quickly critical findings are addressed — not merely whether scans occurred or how many findings were identified. MTTR for critical findings is the metric that demonstrates the program translates findings into reduced organizational risk. A is wrong — total finding count measures coverage but not what happens to those findings; high finding counts can indicate either better scanning or degraded remediation. C is wrong — license count is an input metric measuring tool deployment, not program effectiveness. D is wrong — asset coverage is a useful coverage metric but does not measure what happens after findings are identified.

Q9. Which approach BEST calibrates an AI vulnerability management system's false positive and false negative rates over time?

- A) Set confidence thresholds at deployment and do not adjust them to maintain consistent alert baselines
- B) Measure false positive rates through analyst review feedback and false negative rates through periodic red team assessments, adjusting thresholds accordingly
- C) Disable low-confidence findings entirely to eliminate false positives and focus only on high-confidence detections
- D) Compare results against a competitor scanner and adopt whichever produces fewer total findings

Answer: B **Explanation:** B is correct. NIST AI RMF 1.0 MEASURE function requires ongoing performance tracking and calibration of AI security systems. False positive rates are measured through analyst confirmation of findings; false negative rates are measured by evaluating whether red team-confirmed vulnerabilities were detected by the scanner. Both feedback loops inform threshold adjustment. A is wrong — AI model performance drifts as the threat landscape and organizational environment change; static thresholds become miscalibrated over time. C is wrong — disabling low-confidence findings eliminates false positives but systematically increases false negatives, which is an undocumented risk acceptance decision. D is wrong — fewer total findings could indicate better precision or could indicate a less sensitive tool missing real vulnerabilities.

Q10. An organization's AI-driven vulnerability platform flags a CVSS 5.3 information disclosure vulnerability as its highest-priority finding this week. The analyst is skeptical because the score seems low. Which factors would MOST likely explain this prioritization?

- A) The AI platform has a configuration error that is incorrectly inflating low-severity findings
- B) The affected asset is internet-facing, the vulnerability appears on the CISA KEV catalog with active ransomware exploitation, and the exposed information includes authentication credentials
- C) The platform's ML model has been poisoned by adversarial training data to deprioritize critical findings
- D) CVSS v4.0 recalculated the score from a higher value, and the platform is using the updated scoring

Answer: B **Explanation:** B is correct. Risk-based prioritization combines CVSS with organizational context. A CVSS 5.3 finding can legitimately become the highest-priority item when: the asset is internet-facing (high network exposure), it appears on the CISA KEV catalog (confirmed active exploitation), and the disclosed information enables further attack (authentication credentials). The combination of these three factors elevates the organizational risk well above what CVSS alone reflects. A is wrong — the scenario describes exactly how risk-based prioritization is supposed to work; this is the expected correct behavior of the system. C is wrong — training data poisoning is a sophisticated attack that is unlikely in this scenario and would not systematically affect only one finding. D is wrong — CVSS scores for existing CVEs are occasionally revised, but this would not explain prioritization above higher-scored findings without additional context.